

(a) Data foundation



Historical North Sea metocean data

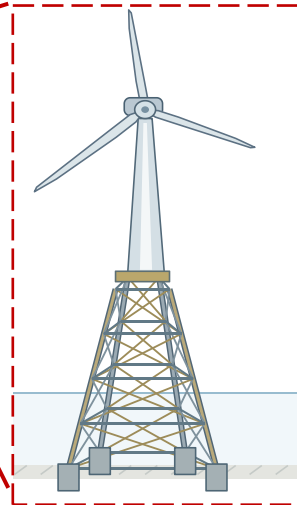
North Sea reanalysis

Historical environmental samples

Fitting
data

Out-of-time
data

(b) Engineering modeling



OC4 structural schematic

Fitting-period
records

Low-fidelity
Analysis

Short-duration
screening

OpenFAST
Simulations

Multifidelity
Response Surrogate

Response-specific
regression

Surrogate-OpenFAST
comparison

Reliability
Metamodel

(c) QAE-ABC optimization

Probability
prediction

QAE amplitude
simulation

$|0\rangle$ Amplitude
simulation $|0\rangle$

Interval Ranking
Design Update

ABC candidate

$\min M(d) \text{ s.t. } P(d) \leq p$